SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
UPS
NDN
AMPK
PROX1
PINK1
PRKN
UPS
nucleus
DNA damage
PARP1
SIRT1
SNCA
S129
ZNF746
GSK3B
NAD depletion
PPARGC1A
T295 S538
T299 T177
MAPK p38
KAT2A
ESRRA
mTORC1
MEF2D
autophagy
CDK5
mitochondrial biogenesis
PPARA
TF PPARA
ESRRA
TF ERRa
YY1
TF YY1
GABPA
TF NRF2
NRF1
TF NRF1
MEF2D
TF MEF2D
MEF2C
TF MEF2C
PPARGC1A
fatty acid oxidation
SIRT3
COX5A
CYCS
COX4I1
COX5A
COX5B
COX6A1
COX6B1
COX6C
COX7A2
COX7B
COX7C
COX8A
CYCS
SDHB
SDHC
SDHD
ATP5F1B
ATP5F1C
POLRMT
TFAM
TFB2M
TFB1M
mitochondrial transcription
SURF1
VDAC1
TOMM20
mitochondrial transport
complex IV
complex II
complex V
mitochondrial biogenesis
RNS
NO
MEF2C
ROS
mutated A53T
SNCA
S129
CHCHD2
TF CHCHD2
hypoxia
ROS
CHCHD4
CHCHD2
CCCP
ROS
PARK7
peroxynitrite
ROS
SIRT3
PINK1
NDUFA9
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
mutated A53T
SNCA
superoxide
H+
SOD1
CHCHD2
mitochondrial depolarization
membrane permeability
purine nucleotide
superoxide
UQCRH
UQCRFS1P1
UQCRQ
UQCR10
UQCR11
UQCRC1
UQCRC2
UQCRFS1
MT-CYB
iron sulfur cluster
heme b
iron sulfur subunit
MT-CYB:heme b
complex III
ubiquinol
rotenone
superoxide
CHCHD2:CHCHD10:CYCS
MT-CO1
MT-CO2
MT-CO3
complex IV
ATP5IF1
ATP5MC1
ATP5MC2
ATP5MC3
F(0)
F(1)
complex V
tiopronin
diazoxide
ATP
ATP-sensitive K+ channel
ROS
mitochondrial dysfunction
SIRT5
myxothiazol
UCP2
UCP3
superoxide
Ca2+
OXPHOS assembly factors
superoxide
superoxide
SDHB
SDHA
Fe-S cluster
FAD
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
ubiquinol
succinic acid
TCA cycle
fumaric acid
SDHB
SDHA
Fe-S cluster
FADH2
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
PARK7
AARS2
DARS2
GARS
KARS
RARS2
Mt tRNA synthetases
POLG2
POLG
MTFMT
MTPAP
Mt replication
mtDNA damage
mtDNA replication
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
mtDNA repair
POLG
ERCC6
ERCC8
Mt-DNA repair
mitochondrial translation
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
mitochondrial transcription
TFAM
nucleoid
NDUFAF1
NDUFAF6
NDUFAF2
NDUFAF5
NDUFAF3
ACAD9
NDUFAF4
FOXRED1
CI
COX10
SCO1
COX14
SCO2
COX15
SURF1
FASTKD2
TACO1
LRPPRC
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
CV
ABCB7
BOLA3
NUBPL
NFU1
Fe-S
OXPHOS assembly factors
CDK5
MAPK14
MAPK12
PARL
HAX1
HAX1:PARL
HTRA2
S400
S212
ROS
PINK1
PINK1
mt UPR
HTRA2
TRAP1
LONP1
CLPB
HSPD1
HSPE1
HSPA9
HSPD1
HSPE1
HSPA9
CLPP
chaperone-mediated protein folding
chaperone-mediated protein folding
proteolysis
mt UPR
mitochondrial respiratory chain
PINK1
PRKN
ribosome
mitochondrion-attached ribosome
PINK1
&
PINK1
&
PINK1
&
mitochondrial matrix
mitochondrion
neuron

AMPK
ribosome
PINK1
ROS
MAPK12
MAPK14
Y182
T180
CDK5
PARL
HAX1
HAX1:PARL
SIRT3
UPS
SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
NDN
PRKN
PINK1
RNS
MEF2C
CHCHD2
TF CHCHD2
GSK3B
ROS
DNA damage
mutated A53T
SNCA
S129
hypoxia
NO
MAPK p38
N:3
S400
S212
HTRA2
peroxynitrite
PARP1
mt UPR
membrane permeability
MEF2C
PROX1
KAT2A
SIRT1
ESRRA
NAD depletion
HSPE1
HSPA9
HSPD1
TRAP1
mitochondrial dysfunction
AARS2
DARS2
GARS
KARS
RARS2
Mt tRNA synthetases
POLG
ERCC6
ERCC8
Mt-DNA repair
TFAM
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
Ca2+
HTRA2
OXPHOS assembly factors
SO2H@C106
PARK7
NDUFA9
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA5
NDUFA6
NDUFA8
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFV1
NDUFV2
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFS2
NDUFS3
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
UQCRH
UQCRFS1P1
UQCRQ
BCS1L
UQCR10
UQCR11
UQCRB
UQCRC1
UQCRC2
UQCRFS1
iron-sulfur cluster
iron-sulfur subunit
MT-CYB
heme b
MT-CYB:heme b
complex III
CLPP
myxothiazol
rotenone
SOD1
T295
S538
T299
T177
PPARGC1A
TF MEF2C
ZNF746
SNCA
S129
CHCHD4
chaperone-mediated protein folding
ATP
Tiopronin
diazoxide
ROS
mtDNA repair
mitochondrial transcription
mitochondrial translation
HSPE1
HSPA9
CLPB
HSPD1
LONP1
neuron
POLG2
POLG
MTFMT
MTPAP
Mt replication
mtDNA damage
chaperone-mediated protein folding
proteolysis
NDUFAF1
NDUFAF6
NDUFAF2
NDUFAF5
NDUFAF3
ACAD9
NDUFAF4
FOXRED1
CI
COX10
SCO1
COX14
SCO2
COX15
SURF1
FASTKD2
TACO1
LRPPRC
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
CV
ABCB7
BOLA3
NUBPL
NFU1
Fe-S
OXPHOS assembly factors
superoxide
superoxide
mTORC1
PPARGC1A
autophagy
CDK5
CHCHD2
PINK1
mitochondrial respiratory chain
KCNJ8
KCNJ11
ATP-sensitive K+ channel
YY1
TF YY1
GABPA
TF NRF2
NRF1
TF NRF1
ESRRA
TF ERRa
PPARA
TF PPARA
MEF2D
TF MEF2D
MEF2D
PRKN
Ub
MT-CO1
MT-CO2
MT-CO3
COX4I1
COX4I2
COX5A
COX5B
COX6A1
COX6A2
COX7A1
COX7A2L
COX7A2
COX7B
COX7C
COX8A
complex IV
mtDNA replication
SLC25A14
mt UPR
SNCA
SLC25A27
mutated A53T
SNCA
SIRT5
SIRT3
ATP5IF1
ATP5MC1
ATP5MC2
ATP5MC3
ATP5PB
ATP5MG
ATP5PD
ATP5MGL
ATP5ME
MT-ATP6
ATP5PF
MT-ATP8
F(0)
ATP5MF
ATP5PO
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
F(1)
complex V
GHITM
CHCHD2
reduced
CYCS
CHCHD2:GHITM:CYCS
UCP3
UCP2
purine nucleotide
CYCS
COX5A
SDHB
SDHC
SDHD
COX4I1
COX7B
COX6A1
COX7A2
COX7C
ATP5F1B
COX6C
COX5B
COX6B1
COX8A
ATP5F1C
TFB1M
POLRMT
TFB2M
TFAM
VDAC1
SURF1
TOMM20
fatty acid oxidation
ROS
SDHB
Fe-S cluster
Fe-S protein
SDHA
FADH2
flavoprotein
SDHC
SDHD
complex II
H+
complex II
complex IV
complex V
mitochondrial transcription
mitochondrial transport
SIRT3
PARK7
SDHB
Fe-S cluster
Fe-S protein
SDHA
FAD
flavoprotein
SDHC
SDHD
complex II
mitochondrial biogenesis
CCCP
H+
fumaric acid
succinic acid
ubiquinol
mitochondrial depolarization
TCA cycle